

HASAN KURBAN

Trustworthy AI & Foundation Model Interpretability · Reasoning & Inference-Time Compute
· *Agentic AI · Multimodal Foundation Models · AI for Science*

Ph.D. in Computer Science, Indiana University Bloomington, 2017

Assistant Professor · College of Science and Engineering
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BIOGRAPHICAL SKETCH. Hasan Kurban directs the *Kurban Intelligence Lab (KIL)* at Hamad Bin Khalifa University, with adjunct appointments at Indiana University Bloomington and Texas A&M University at Qatar. His research investigates the foundations of **trustworthy and interpretable large-scale AI systems**, drawing on methods from machine learning, statistical inference, and the natural sciences. The lab's work spans **trustworthy AI and foundation model interpretability** (hallucination detection and mitigation, sparse-autoencoder-based feature analysis, model editing, uncertainty quantification, red-teaming); **reasoning and inference-time compute** (reinforcement learning with verifiable rewards, abstention-aware reasoning, posterior-constrained inference, multi-agent reasoning); **agentic AI and LLM agents** (tool use, retrieval-augmented and knowledge-grounded reasoning, LLM-as-a-judge evaluation, agent safety, scientific and biomedical discovery agents); **multimodal foundation models** (vision-language models, video understanding and long-context temporal reasoning, world models for physical systems, embodied and spatial reasoning); and **AI for science** (scientific foundation models, equivariant interatomic potentials, physics-informed and symmetry-aware learning, with applications to materials informatics, quantum chemistry, biomedical imaging, and energy systems). Recent work has appeared at *NeurIPS*, *ICML*, *ICLR*, *CVPR*, *ECCV*, *ACL*, *EMNLP*, *COLM* and *KDD*, and in *IEEE Transactions*. He serves on the editorial board of *Nature Scientific Reports* and leads a research group of **15 students** supported by QRDI, TÜBİTAK, and partner research universities.

73 Journal Articles · 68 Conference Papers · 16 Funded Grants · 15 Research Students · 718 Citations · *h-index* 15

RESEARCH INTERESTS

- **Trustworthy AI and Foundation Model Interpretability:** hallucination detection and mitigation in large language models; sparse-autoencoder-based feature analysis; model editing; uncertainty quantification and calibration; jailbreak robustness and red-teaming; auditable behavior in multimodal and scientific systems.
- **Reasoning and Inference-Time Compute:** reinforcement learning with verifiable rewards (RLVR); abstention-aware reasoning; posterior-constrained inference for mathematical reasoning; multi-agent reasoning frameworks; neurosymbolic methods; robustness of chain-of-thought to noisy, conflicting, and adversarial evidence.
- **Agentic AI and LLM Agents:** long-horizon autonomous agents with tool use and planning; retrieval-augmented and knowledge-grounded reasoning; LLM-as-a-judge evaluation; agent safety; multi-agent coordination across heterogeneous knowledge sources; scientific and biomedical discovery agents.
- **Multimodal Foundation Models:** vision-language models; video understanding and long-context temporal reasoning; world models for physical systems; embodied and spatial reasoning; multimodal benchmarks for crystalline materials, medical imaging, and quantum chemistry.
- **AI for Science:** scientific foundation models; equivariant interatomic potentials; physics-informed and symmetry-aware learning; active learning for materials informatics, quantum chemistry, biomedical imaging, and energy systems.

RESEARCH FUNDING AND GRANTS

Competitive Research Grants (PI / Lead PI)

- 2026 **QRDI: Agentic Artificial Intelligence Grant.** A Governance and Audit Toolkit for the Responsible Adoption of Agentic Artificial Intelligence in Qatar's Financial Sector.
Role: Lead Principal Investigator • Award No. AAICo1-0130-260007 • Budget: \$41,000.
- 2026 **QRDI: Agentic Artificial Intelligence Grant.** Agentic AI Assurance for Qatar: Global Governance Mapping, GCC Benchmarking, and a Finance-Focused Policy Toolkit.
Role: Principal Investigator • Award No. AAICo1-0208-260013 • Budget: \$40,978.
- 2025 **QRDI: Modernizing Charitable Sector Grant.** Advancing AI-Driven Governance in Qatar's Charitable Sector: A Trustworthy Arabic LLM for Regulatory Compliance.
Role: Lead Principal Investigator • Award No. MCSCo2-0221-250019 • Budget: \$40,999.
- 2025 **QRDI: Undergraduate Research Experience Program (UREP).** Leveraging Data Analytics for Interpreting Transformer Dissolved Gas Analysis.
Role: Principal Investigator • Award No. UREP31-043-2-014 • Budget: \$26,042.
- 2025 **TÜBİTAK 1001 Scientific and Technological Research Projects Support Program.** Improving Dynamic Fit and Comfort of Prosthetic Sockets Using Wearable Fiber Optic Sensors.
Role: Principal Investigator • Award No. 125E223 • Budget: \$60,000.

- 2023 **QRDI: National Priorities Research Program (NPRP).** ReSolVE: Resilient Solutions for Vulnerabilities and Emergencies: An Effective National Risk Management Plan for Qatar.
Role: Principal Investigator (joined 2025) • Award No. NPRP14C-0909-210008 • Budget: \$4,434,601.
- 2021 **QRDI: National Priorities Research Program Cluster (NPRP-C).** Multi-layer Cybersecurity and Situational Awareness to Enhance Resiliency in Qatar's Power Grid.
Role: Principal Investigator (joined 2026) • Award No. NPRP12C-0814-190012 • Budget: \$3,204,904.

Institutional Grants and Internal Funding

- 2026 **Student Research Experience Grant (SREG),** Texas A&M University at Qatar. Physics-Guided 4D Gaussian Splatting for Sparse-View Dynamic Scene Understanding.
Role: Principal Investigator • Budget: \$44,000.
- 2025 **HBKU Horizon Hub 1st Cycle Program,** Office of the VP for Research. AI for Science Symposium (January 21–22, 2026).
Role: Principal Investigator • Award No. HBKU-INT-VPR-HHW-01-8 • Budget: QAR 150,000.
- 2025 **HBKU Thematic Research Grant Program (3rd Cycle).** Optimal AI and Quantum-Secured Wireless–Optical Communication Design for Smart Cities in Qatar.
Role: Principal Investigator • Award No. HBKU-INT-VPR-TG-03-04 • Budget: \$102,740.
- 2025 **HBKU GPU Compute Grant,** Research Computing Core Group. Compressed Motion Statistics for Reliable Evaluation of Generative Videos.
Role: Lead Principal Investigator • Budget: QAR 7,000.
- 2024 **Transformative Educational Experience (TEE) Grant,** Texas A&M University at Qatar. Transforming Software Engineering Education through Autograder and LLM Integration.
Role: Lead Principal Investigator • Budget: \$10,000.
- 2024 **Multiversity Academic Grant,** Texas A&M University at Qatar. A Novel, Multi-Institutional Machine Learning for Engineers Course.
Role: Lead Principal Investigator • Budget: \$10,000.
- 2023 **Faculty Resource Allocation (Three-Year),** Texas A&M University at Qatar.
Role: Lead Principal Investigator • Budget: \$53,750.
- 2023 **Research Impact Initiative,** Texas A&M University at Qatar. Leveraging AI, Machine Learning, and Data Analytics to Enhance Qatar's Sustainable Energy, Healthcare, and Security.
Role: Lead Principal Investigator • Budget: \$177,000.
- 2023 **Startup Research Funds (Three-Year),** Texas A&M University at Qatar.
Role: Lead Principal Investigator • Budget: \$413,000.

SELECTED PUBLICATIONS

Selected recent and high-impact work; complete list of journal articles and conference papers follows. **Bold** denotes Dr. Kurban. A dagger ([†]) indicates papers led by Dr. Kurban as senior supervising author and principal investigator, with the first author being a doctoral, master's, or undergraduate advisee or co-advisee of Dr. Kurban.

- **EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration.**
M. R. Barhdadi, M. Tuncel, E. Serpedin, **H. Kurban[†]**.
The 39th Conf. on Neural Information Processing Systems (NeurIPS), 2025.
- **SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs.**
S. Abdaljalil, F. Pallucchini, A. Seveso, **H. Kurban[†]**, F. Mercorio, E. Serpedin.
30th Conf. on Empirical Methods in Natural Language Processing (EMNLP), 2025.
- **xChemAgents: Agentic AI for Explainable Quantum Chemistry.**
C. Polat, M. Tuncel, M. Kurban, E. Serpedin, **H. Kurban[†]**.
The 42nd Int. Conf. on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models), 2025.
- **Geometric- k -means: A Bound-Free Approach to Fast and Eco-Friendly k -means.**
P. Sharma, M. Malec, **H. Kurban[†]**, O. Kulekci, M. M. Dalkilic.
Machine Learning, 2025.
- **p -ClustVal: A Novel p -adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data.**
P. Sharma, S. Mishra, **H. Kurban[†]**, M. M. Dalkilic.
International Journal of Data Science and Analytics, 2025.
- **Rapidly Predicting Kohn–Sham Total Energy Using Data-centric AI.**
H. Kurban[†], M. Kurban, M. M. Dalkilic.
Nature Scientific Reports, 12, 1–14, 2022.
- **EM*: An EM Algorithm for Big Data.** HONORABLE MENTION AWARD
H. Kurban[†], M. Jenne, M. M. Dalkilic.
IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2016.

PUBLICATIONS

Peer-Reviewed Journal Articles

Items marked (*under review*) at the right margin are currently under peer review.

- [73] C. Polat, E. Serpedin, M. Kurban, **H. Kurban[†]**. ‘Grounded verification of chemical and materials reasoning: detection is the bottleneck.’ *Nature Machine Intelligence*, 2026.
(*under review*)
- [72] M. Kurban, C. Polat, E. Serpedin, **H. Kurban[†]**. “Local motif-guided discovery of hydrogen-stabilizing environments in refractory high-entropy alloys” submitted.” *Journal of Alloys and Compounds*, 2026.
(*under review*)
- [71] **H. Kurban**, M. Kurban. “Certified discovery: statistically valid materials screening claims

- from a small DFT verification budget. " *Nature Communications*, 2026. (under review)
- [70] **H. Kurban**, M. Kurban. "Materials foundation models fail together: correlated errors and collective blind spots. " *npj Computational Materials*, 2026. (under review)
- [69] V. Tataroglu, **H. Kurban**[†]. "The Persuasion–Prediction Gap: Outcome-Grounded Auditing of LLM Judges on Market-Resolved Financial and Crypto Reasoning." *Expert Systems with Applications*, 2026. (under review)
- [68] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. "Site-engineered reversible H₂ adsorption on Zr-doped anatase TiO₂ nanoparticles: active learning-assisted hydrogen storage screening." *International Journal of Hydrogen Energy*, 2026. (under review)
- [67] S. Abdaljalil, E. Serpedin, K. Qaraqe, **H. Kurban**[†]. "Audit-of-Understanding: Posterior-Constrained Inference for Mathematical Reasoning in Language Models," *Artificial Intelligence*, 2026. (under review)
- [66] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine-Robust, Privacy-Preserving Distributed Unit Commitment Using a Hybrid Approach," *IEEE Transactions on Automation Science and Engineering*, 2026. (under review)
- [65] M. M. Islam, M. Ismail, A. Takiddin, **H. Kurban**, Rachad Atat, E. Serpedin. "Evaluating False Data Injection Detectors in Smart Grid Under Stealthy, High-Impact Attacks," *IEEE Transactions on Smart Grid*, 2026. (under review)
- [64] M. R. Islam, S. Abdaljalil, E. Serpedin, **H. Kurban**[†]. "ConfTriage: Calibration-Aware Large Language Model Triage for Pulmonary Nodule Malignancy with Selective Specialist Deferral," *IEEE Journal of Biomedical and Health Informatics*, 2026. (under review)
- [63] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**[†]. "SCALAR: Quantifying Structural Hallucination, Consistency, and Reasoning Gaps in Materials Foundation Models." *Digital Discovery*, 2026. (under review)
- [62] C. Polat, M. Kurban, E. Serpedin, **H. Kurban**[†]. "QuantumCanvas: A Multimodal Benchmark for Learning Two-Body Quantum Interactions." *Machine Learning: Science and Technology*, 2026. (under review)
- [61] R. Khanbayov, **H. Kurban**[†]. "Topical Phase Transitions in Artificial Intelligence Research: Large-Scale Evidence and an Early-Warning Signature for Emerging Topics," *Quantitative Science Studies*, 2026. (under review)
- [60] M. Elnour, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. "Robust Graph Contrastive Learning for Attack Detection in Power Distribution Systems," *IEEE Open Access Journal of Power and Energy*, 2026. (under review)
- [59] M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. "Enhancing Robustness of FDI Attack Detectors in Smart Grids Against Adaptive Adversaries via a Physics-Constrained GAN," *Electric Power System Research*, 2026. (under review)
- [58] M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. "Byzantine-Resilient Distributed EV Charging Using Threshold Cryptography and Machine Learning," *IEEE Transactions on Transportation Electrification*, 2026. (under review)
- [57] C. Polat, M. Kurban, E. Serpedin, **H. Kurban**[†]. "CliffordIP: Clifford Algebra Equivariant Interatomic Potentials for Heterogeneous Catalysis," *npj Computational Materials*, 2026.

(under review)

- [56] M. Kurban, E. Serpedin, **H. Kurban**. "Sequential Molecular Hydrogen Loading on the ZIF-8 Metal–Organic Framework: Adsorption Hierarchy, Electronic Structure, and Charge Transfer Mechanism," *International Journal of Hydrogen Energy*, 2026. (under review)
- [55] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. "Physics-Guided Thermodynamic Screening of H_2 Adsorption on Alkaline-Earth-Modified Anatase TiO_2 Nanoparticles with LLM-Assisted Interpretation," *RSC Advances*, 2026. (under review)
- [54] M. Elnour, M. AlShaikh Saleh, M. O. Cakiroglu, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. "A Deep-Learning Based Approach for Photovoltaic Forecasting Using Symbolic Graph Representations," *IEEE Open Access Journal of Power and Energy*, 2026. (under review)
- [53] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine Fault-Tolerant Lightweight Security Framework for Distributed DC Optimal Power Flow Using Threshold Cryptography," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. (under review)
- [52] S. Abdaljalil, E. Serpedin, **H. Kurban**[†]. "HalluVerse3: A Fine-Grained Multilingual Benchmark for Hallucination Detection in LLMs," *Language Resources and Evaluation*, 2026. (under review)
- [51] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine-Resilient, Privacy-Preserving, Scalable Distributed Unit Commitment Using Threshold Cryptography," *IEEE Open Journal of Power Systems Society*, 2026. (under review)
- [50] M. Elnour, M. AlShaikh Saleh, R. Atat, X. Huo, A. Takiddin, M. Ismail, **H. Kurban**, K. R. Davis, E. Serpedin. "Joint Sensor Deployment and Physics-Informed Graph Transformer for Smart Grid Attack Detection," *IEEE Open Access Journal of Power and Energy*, 2026. (under review)
- [49] M. R. Islam, M. Y. Ansari, **H. Kurban**[†], E. Serpedin. "A Multimodal Interpretable Approach to Lung Nodule Diagnosis from CT with Radiology and Anatomy-Aware Text," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2026. (under review)
- [48] I. B. Altun, M. O. Cakiroglu, S. Awadallah, M. M. Dalkilic, **H. Kurban**[†]. "Benchmarking Artificial Intelligence Models for Dissolved Gas Forecasting in Power Transformers," *Engineering Applications of Artificial Intelligence*, 2026. (under review)
- [47] M. R. Islam, M. M. Islam, M. Y. Ansari, **H. Kurban**[†], E. Serpedin. "Agentic AI in Biomedical and Healthcare: A Modality-Centric Review of Emerging Frameworks, Open Challenges, and Future Directions," *Artificial Intelligence Review*, 2026.
- [46] M. Kurban, E. Serpedin, **H. Kurban**. "Hydrogen Storage in TiFe Nanoparticles: Void-Limited Saturation and Metal-Selective Charge Transfer During Sequential Loading," *International Journal of Hydrogen Energy*, 2026.
- [45] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. "Explainable Machine Learning Reveals Orbital Design Rules for Reversible H_2 Adsorption on Single-Atom-Doped TiO_2 Nanoparticles," *Materials & Design*, 2026.
- [44] M. M. Islam, M. Ismail, **H. Kurban**, X. Huo, E. Serpedin. "Lightweight Defense Against Data Consistency Attacks in Distributed DC Optimal Power Flow," *IEEE Transactions on*

Automation Science and Engineering, 2026.

- [43] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. “Descriptor-guided Thermodynamic Screening of H_2 Adsorption on Single-atom-doped Anatase TiO_2 Nanoparticles with Interpretable Machine Learning,” *Scientific Reports*, 2026.
- [42] M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin. “Linear-Complexity Unified Defense Against Deception Attacks in Distributed Economic Dispatch Using Cryptography and Machine Learning,” *IEEE Transactions on Smart Grid*, 2026.
- [41] M. A. Yatkin, M. Körgesaar, V. M. Asan, J. Romanoff, J. Stuckner, **H. Kurban**. “Self-Consistent Recurrent Neural Network for Path Dependent Deformation,” *Scientific Reports*, 2026.
- [40] S. Abdaljalil, P. Sharma, R. Atat, E. Serpedin, **H. Kurban**[†]. “SINdex: Semantic INconsistency Index for Hallucination Detection in LLMs,” *IEEE Open Journal of the Computer Society*, 2026.
- [39] M. R. Islam, M. K. Hasan, **H. Kurban**[†], E. Serpedin. “Bayesian Probabilistic Knowledge from Diameter Prior for Decision Fusion to Detect Lung Nodule Heterogeneity,” *IEEE Transactions on Artificial Intelligence*, 2026.
- [38] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. “Multimodal Explainable AI-Driven Analysis of Quantum Size Effects in Copper Nanoclusters for Hydrogen Storage,” *International Journal of Hydrogen Energy*, 2026.
- [37] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. “Engineered MgO Nanoparticles with Tunable Electronic Signatures for Energy Applications,” *ACS Applied Nano Materials*, 2026.
- [36] H. Cetinkaya, F. Ay, M. Tuncel, H. Nounou, M. N. Nounou, **H. Kurban**, E. Serpedin. “Curriculum-Enhanced Adaptive Sampling for Physics-Informed Neural Networks: A Robust Framework for Stiff PDEs,” *Mathematics*, 2025.
- [35] **H. Kurban**, P. Sharma, M. Dalkilic, M. Kurban. “Accelerating Density of States Prediction in Zn-Doped MgO Nanoparticles via Kernel-Optimized Weighted k -NN,” *Scientific Reports*, 2025.
- [34] S. Abdaljalil, **H. Kurban**[†], R. Atat, E. Serpedin, K. Qaraqe. “Deep Temporal and Structural Embeddings for Robust Unsupervised Anomaly Detection in Dynamic Graphs,” *IEEE Open Journal of the Computer Society*, 2025.
- [33] M. K. Ramachandra, **H. Kurban**[†], M. O. Kulekci, M. M. Dalkilic. “Telescope Indexing for k -Nearest Neighbor Search Algorithms over High-Dimensional Data & Large Data Sets,” *Scientific Reports*, 2025.
- [32] P. Sharma, M. Malec, **H. Kurban**[†], O. Kulekci, M. M. Dalkilic. “Geometric- k -means: A Bound-Free Approach to Fast and Eco-Friendly k -means,” *Machine Learning*, 2025.
- [31] M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**[†], M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin. “An Extended Frequency-Improved Legendre Memory Model for Enhanced Long-term Electricity Load Forecasting,” *IEEE Open Access Journal of Power and Energy*, 2025.
- [30] C. Polat, **H. Kurban**[†], M. Kurban. “Enabling Ease of Access to Quantum Chemistry with Transformer-Based Text Encoding and Physics-Informed Multilayer Perceptron,” *Physica*

- Scripta*, 2025.
- [29] M. O. Cakiroglu, **H. Kurban**[†], E. K. Buxton, M. Dalkilic. "A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting using TimesNet," *IEEE Access*, 2025.
 - [28] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**[†]. "CrysMTM: A Multiphase, Temperature-Resolved, Multimodal Dataset for Crystalline Materials," *Machine Learning: Science and Technology*, 2025.
 - [27] C. Polat, **H. Kurban**[†], M. Kurban. "QuantumShellNet: Ground-State Eigenvalue Prediction of Materials Using Electronic Shell Structures and Fermionic Properties via Convolutions," *Computational Materials Science*, 246, 113366, 2025.
 - [26] P. Sharma, S. Mishra, **H. Kurban**[†], M. M. Dalkilic. "*p*-ClustVal: A Novel *p*-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data," *International Journal of Data Science and Analytics*, 2025.
 - [25] F. Ay, S. Althunibat, K. Qaraqe, **H. Kurban**[†]. "A Noise-Adaptive Machine Learning Framework for Optimizing User Grouping in Dynamic IM-OFDMA Systems," *IEEE Transactions on Communications*, 2024.
 - [24] M. O. Cakiroglu, **H. Kurban**[†], K. Qaraqe, L. Aljihmani, G. Petrovski, M. M. Dalkilic. "A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph," *Scientific Reports*, 2024.
 - [23] C. Polat, M. Kurban, **H. Kurban**[†]. "Multimodal Neural Network-Based Predictive Modeling of Nanoparticle Properties from Pure Compounds," *Machine Learning: Science and Technology*, 2024.
 - [22] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†]. "Enhancing the Electronic Properties of TiO₂ Nanoparticles Through Carbon Doping: An Integrated DFTB and Computer Vision Approach," *Computational Materials Science*, 244, 113248, 2024.
 - [21] M. O. Cakiroglu, **H. Kurban**[†], P. Sharma, M. O. Kulekci, E. K. Buxton, M. Raeeszadeh-Sarmazdeh, M. M. Dalkilic. "An Extended de Bruijn Graph for Feature Engineering Over Biological Sequential Data," *Machine Learning: Science and Technology*, 5(3), 035020, 2024.
 - [20] S. Temiz, S. Erol, **H. Kurban**[†], M. M. Dalkilic. "State of Charge and Temperature-Dependent Impedance Spectra Regeneration of Lithium-Ion Battery by Duplex Learning Modeling," *Journal of Energy Storage*, 64, 107085, 2023.
 - [19] **H. Kurban**, M. Kurban, M. M. Dalkilic. "Rapidly Predicting Kohn–Sham Total Energy Using Data-centric AI," *Nature Scientific Reports*, 12, 1–14, 2022.
 - [18] S. Temiz, **H. Kurban**, S. Erol, M. M. Dalkilic. "Data on Machine Learning Regenerated Lithium-ion Battery Impedance," *Data in Brief*, 108698, 2022.
 - [17] M. S. Malec, **H. Kurban**, M. M. Dalkilic. "ccImpute: An Accurate and Scalable Consensus Clustering Based Algorithm to Impute Dropout Events in Single-Cell RNA-seq Data," *BMC Methods*, 23, 1–17, 2022.
 - [16] S. Temiz, **H. Kurban**, S. Erol, M. M. Dalkilic. "Regeneration of Lithium-ion Battery Impedance Using a Novel Machine Learning Framework and Minimal Empirical Data," *Journal of Energy Storage*, 52, 105022, 2022.

- [15] P. Sharma, **H. Kurban**, M. M. Dalkilic. "DCEM: An R Package for Clustering Big Data via Data-centric Modification of Expectation Maximization," *SoftwareX*, 17, 100944, 2022.
- [14] **H. Kurban**, M. Kurban. "Building Machine Learning Systems for Multi-Atoms Structures: CH₃NH₃PbI₃ Perovskite Nanoparticles," *Computational Materials Science*, 195, 110490, 2021.
- [13] **H. Kurban**, M. Kurban, P. Sharma, M. M. Dalkilic. "Predicting Atom Types of Anatase TiO₂ Nanoparticles with Machine Learning," *Key Engineering Materials*, 880, 89–94, 2021.
- [12] **H. Kurban**, M. Kurban. "Rare-class Learning over Mg-Doped ZnO Nanoparticles," *Chemical Physics*, 546, 11159, 2021.
- [11] I. Muz, **H. Kurban**, M. Kurban. "A DFT Study on Stability and Electronic Structure of AlN Nanotubes," *Materials Today Communications*, 26, 102118, 2021.
- [10] **H. Kurban**, S. Alaei, M. Kurban. "Effect of Mg Content on Electronic Structure, Optical and Structural Properties of Amorphous ZnO Nanoparticles: A DFTB Study," *Journal of Non-Crystalline Solids*, 560, 120726, 2021.
- [9] **H. Kurban**[†]. "Atom Classification with Machine Learning and Correlations among Physical Properties of ZnO Nanoparticle," *Chemical Physics*, 545, 111143, 2021.
- [8] **H. Kurban**[†]. "Measuring the Proximity of Medical Treatment Areas with Text Mining," *European Journal of Science and Technology*, 21, 518–526, 2021.
- [7] **H. Kurban**[†]. "Practical Data Science: Examining the Correlations between Structural and Electronic Properties of Different Phases of TiO₂ Nanoparticles," *Journal of Selcuk-Technic*, 4(19), 1–9, 2020.
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- [5] **H. Kurban**, M. Kurban. "Study of Structural and Optoelectronic Properties of Hexagonal ZnO Nanoparticles," *Bilecik Seyh Edebali University Journal of Science*, 6(2), 124–131, 2019.
- [4] M. Kurban, **H. Kurban**, M. M. Dalkilic. "Controlling Structural and Electronic Properties of ZnO NPs," *Bilge International Journal of Science and Technology Research*, 3(0), 35–39, 2019.
- [3] **H. Kurban**, M. Kurban, M. M. Dalkilic. "Density-functional Tight-binding Approach for the Structural Analysis and Electronic Structure of Copper Hydride Metallic Nanoparticles," *Materials Today Communications*, 21, 100648, 2019.
- [2] **H. Kurban**, M. Jenne, M. M. Dalkilic. "Using Data to Build a Better EM: EM* for Big Data," *International Journal of Data Science and Analytics*, 4(2), 83–97, 2017.
- [1] M. Jenne, O. Boberg, **H. Kurban**, M. M. Dalkilic. "Studying the Milky Way Galaxy using ParaHeap-k, a Parallel Heap-based *k*-means," *IEEE Computer*, 47(9), 26–33, 2014.

Peer-Reviewed Conference Proceedings

Items marked (*under review*) at the right margin are currently under peer review.

- [68] **It's the Decoding Format, Not the Perturbation: Auditing Consistency-Based Selection for Vision-Language Test-Time Scaling**
P. Zheng, **H. Kurban**[†].
41th AAAI, Singapore, 2027. (*under review*)

- [67] **A Copy Can Beat Deep Models: When Retrieval Helps Time-Series Forecasting**
M. O. Cakiroglu, E. K. Buxton, M. Dalkilic, **H. Kurban[†]**.
41th AAI, Singapore, 2027. (under review)
- [66] **Detect Early, Escalate Rarely: Anytime Detection of AI-Generated Video from the Compressed Bitstream**
M. O. Cakiroglu, M. Dalkilic, **H. Kurban[†]**.
41th AAI, Singapore, 2027. (under review)
- [65] **Conformal Coverage Guarantees for Any Video Temporal Grounder**
A. Mohamed, R. Khanbayov, E. Serpedin, **H. Kurban[†]**.
41th AAI, Singapore, 2027. (under review)
- [64] **The Spectrum Is Not Enough: When Context Helps Time-Series Forecasting**
M. O. Cakiroglu, M. Mehmet Dalkilic, **H. Kurban[†]**.
41th AAI, Singapore, 2027. (under review)
- [63] **4D Synchronized Fields: Motion-Language Gaussian Splatting.**
M. R. Barhdadi, S. Abdaljalil, R. Khanbayov, E. Serpedin, **H. Kurban[†]**.
41th AAI, Singapore, 2027. (under review)
- [62] **A Cost-Aware, Paired Protocol for Auditing Dynamic Tool Synthesis in Agentic Video Question Answering.**
A. Mohamed, R. AlHamidi, M. R. Barhdadi, R. Khanbayov, E. Serpedin, **H. Kurban[†]**.
27th IEEE/CVF WACV, Buena Vista, FL, USA, 2027. (under review)
- [61] **LGQ: Learnable Geometric Quantization for Image Tokenization.**
I. B. Altun, M. O. Cakiroglu, E. K. Buxton, M. Dalkilic, **H. Kurban[†]**.
27th IEEE/CVF WACV, Buena Vista, FL, USA, 2027. (under review)
- [60] **Auditing Generalization in AI-Generated Video Detection: A Six-Control Protocol and the VidAudit Toolkit?**
M. O. Cakiroglu, Z. Lu, M. Mehmet Dalkilic, **H. Kurban[†]**.
27th IEEE/CVF WACV, Buena Vista, FL, USA, 2027. (under review)
- [59] **Edge-Conditioned Contrastive Graph Learning for False Data Injection Detection in Transmission Power Systems.**
M. Elnour, R. Atat, , **H. Kurban**, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin.
52nd IECON, Doha, Qatar, 2026. (under review)
- [58] **Hydrogen-Rich Surfaces Enhance Storage Capacity in MgAlH₃ Perovskite Nanoparticles.**
M. Kurban, E. Serpedin, **H. Kurban**.
52nd IECON, Doha, Qatar, 2026. (under review)
- [57] **Rank Reversal in Multilingual LLM Judges: A Label-Free Double-Centering Calibrator.**
A. S. Mahmood, S. Abdaljalil, **H. Kurban[†]**.
31st EMNLP, Budapest, Hungary, 2026. (under review)
- [56] **Correct but Not Grounded: Stress Testing Evidence Utilization in Large Language Models.**
S. Abdaljalil, E. Serpedin, **H. Kurban[†]**.
31st EMNLP, Budapest, Hungary, 2026. (under review)

- [55] **Knowing When Not to Answer: Abstention-Aware Scientific Reasoning.**
S. Abdaljalil, E. Serpedin, **H. Kurban[†]**.
31st EMNLP, Budapest, Hungary, 2026. (under review)
- [54] **Intrinsic 4D Gaussian Segmentation from Scene Cues.**
Hasan Yazar, Mohamed Rayan Barhdadi, Erchin Serpedin, Mehmet Tuncel, **H. Kurban[†]**.
37th BMVC, Lancaster, UK, 2026. (under review)
- [53] **Halluverse-M³: A Multitask Multilingual Benchmark for Hallucination in LLMs.**
S. Abdaljalil, P. Sharma, E. Serpedin, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [52] **Perception, Not Reasoning, Limits Visual Theory of Mind.**
M. R. Barhdadi, S. T. Wasim, J. Gall, E. Serpedin, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [51] **STEMGym: A Gymnasium Environment for Benchmarking Dose-Efficient Autonomous Scanning Transmission Electron Microscopy.**
C. Polat, E. Serpedin, M. Kurban, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [50] **CliffordSTF: Constructive Clebsch-Gordan Completeness Without Spherical Harmonics.**
C. Polat, E. Serpedin, M. Kurban, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [49] **GRAPE: Graph-Augmented Prototype Explanations for Interactive Medical Image Diagnosis.**
R. Khanbayov, E. Serpedin, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [48] **IsoSci: A Benchmark of Isomorphic Cross-Domain Science Problems for Evaluating Reasoning versus Knowledge Retrieval in LLMs.**
S. Abdaljalil, E. Serpedin, **H. Kurban[†]**.
40th NeurIPS, Sydney, Australia, 2026. (under review)
- [47] **Physics-Informed Data Augmentation and Explainable Machine Learning for Hydrogen Storage Prediction in LiNH₆ Nanoparticles.**
M. Kurban, E. Serpedin, **H. Kurban**.
52nd IECON, Doha, Qatar, 2026.
- [46] **Machine Learning-Based Robust Distributed Economic Dispatch Under Data Integrity Attacks.**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
17th IEEE SmartGridComm, College Station, TX, United States, 2026.
- [45] **IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video.**
R. Khanbayov, M. R. Barhdadi, E. Serpedin, **H. Kurban[†]**.
19th ECCV, Malmö, Sweden, 2026.
- [44] **Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation.**

- A. S. Mahmood, S. Abdaljalil, **H. Kurban**[†].
3rd COLM, San Francisco, CA, USA, 2026.
- [43] **Perception, Not Reasoning, Limits Visual Theory-of-Mind in Multimodal Foundation Models.**
M. R. Barhdadi, E. Serpedin, **H. Kurban**[†].
43rd IEEE/CVF CVPR, Multi-Modal Reasoning for Agentic Intelligence, Denver, CO, USA, 2026.
- [42] **How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science.**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**[†].
32nd ACM SIGKDD KDD, Jeju, Korea, 2026.
- [41] **Evaluating Trustworthy Multilingual Alignment in LLMs via Synthetic Natural Language Inference.**
S. Abdaljalil, K. Qaraqe, E. Serpedin, **H. Kurban**[†].
35th IJCAI & 29th ECAI, Trust-AI, Bremen, Germany, 2026.
- [40] **Monte Carlo Dropout for Robust and Uncertainty-Aware Lung Nodule Classification.**
M. R. Islam, C. Mondal, M. Y. Ansari, **H. Kurban**[†], E. Serpedin.
34th EUSIPCO, Bruges, Belgium, 2026.
- [39] **Fair-FL: Verifiable Inclusive Aggregation for Fair Federated Learning.**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
34th EUSIPCO, Bruges, Belgium, 2026.
- [38] **Strategic Bidding in Competitive and Adversarial Electricity Markets Using No-Regret Learning Algorithms.**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
23rd IFAC World Congress, Busan, Korea, 2026.
- [37] **Site Selectivity Controls Reversible Hydrogen Adsorption and Release on TiFe Nanoparticles: An Interpretable Machine Learning-Assisted Study.**
M. Kurban, E. Serpedin, **H. Kurban**[†].
6th International Conf. on Electrical, Computer and Energy Technologies (ICECET), Rome, Italy, 2026.
- [36] **Agentic and Explainable Intelligent Screening of Metal–Organic Frameworks for CO₂ Direct Air Capture.**
M. Kurban, E. Serpedin, M. Ergun, **H. Kurban**[†].
8th Intelligent and Fuzzy Systems Conf. (INFUS), Ankara, Turkey, 2026.
- [35] **Thermodynamic and Electronic Mapping for Hydrogen Storage on Zr-Decorated TiO₂ Nanoparticles with an Interpretable Learning Agent and Computational Modeling.**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**[†].
25th International Conf. on Advanced Nanomaterials (ANM), Aveiro, Portugal, 2026.
- [34] **EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration.**
M. R. Barhdadi, M. Tuncel, E. Serpedin, **H. Kurban**[†].
39th Conf. on Neural Information Processing Systems (NeurIPS), San Diego, CA, USA, 2025.
- [33] **Learning Physics Like Humans: Uncovering Physical Laws from Monocular Videos.**

- M. R. Barhdadi, H. Alnuweiri, **H. Kurban**[†].
21st International Conf. on Computer Vision (ICCV, Human-inspired Computer Vision), Hawaii, USA, 2025.
- [32] **SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs.**
S. Abdaljalil, F. Pallucchini, A. Seveso, **H. Kurban**[†], F. Mercurio, E. Serpedin.
30th Conf. on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China, 2025.
- [31] **Stress-Testing Multimodal Foundation Models for Crystallographic Reasoning.**
C. Polat, **H. Kurban**[†], E. Serpedin, M. Kurban.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models), Vienna, Austria, 2025.
- [30] **Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models.**
S. Abdaljalil, **H. Kurban**[†], K. Qaraqe, E. Serpedin.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models), Vienna, Austria, 2025.
- [29] **xChemAgents: Agentic AI for Explainable Quantum Chemistry.**
C. Polat, M. Tuncel, M. Kurban, E. Serpedin, **H. Kurban**[†].
42nd International Conf. on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models), Vancouver, Canada, 2025.
- [28] **Beyond Atomic Geometry Representations in Materials Science: A Human-in-the-Loop Multimodal Framework.**
C. Polat, **H. Kurban**[†], E. Serpedin, M. Kurban.
42nd International Conf. on Machine Learning (ICML, DataWorld), Vancouver, Canada, 2025.
- [27] **PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views.**
M. R. Barhdadi, **H. Kurban**[†], H. Alnuweiri.
42nd International Conf. on Machine Learning (ICML, Building Physically Plausible World Models), Vancouver, Canada, 2025.
- [26] **Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting.**
M. O. Cakiroglu, I. B. Altun, M. Dalkilic, E. K. Buxton, **H. Kurban**[†].
42nd International Conf. on Machine Learning (ICML, Foundation Models for Structured Data), Vancouver, Canada, 2025.
- [25] **Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding.** SPOTLIGHT
C. Polat, **H. Kurban**[†], E. Serpedin, M. Kurban.
42nd IEEE/CVF Computer Vision and Pattern Recognition Conf. (CVPR, MM4Mat), Nashville, TN, USA, 2025.
- [24] **TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials.**
C. Polat, **H. Kurban**[†], E. Serpedin, M. Kurban.

- 13th International Conf. on Learning Representations (ICLR, AI for Accelerated Materials Design)*, Singapore, 2025.
- [23] **De Bruijn Graph-Enhanced Time Series Models for Electricity Load Forecasting.**
M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**[†], M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin, K. Qaraqe.
17th International Symposium on Signals, Circuits and Systems (SSCS), Iași, Romania, 2025.
- [22] **Data-Efficient Hydrogen Adsorption Prediction in Copper Nanoclusters: A Computer Vision-Based Transfer Learning Approach.**
C. Polat, **H. Kurban**[†], E. Serpedin, M. Kurban.
19th International Conf. on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Turkey, 2025.
- [21] **Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics.**
P. Sharma, **H. Kurban**[†], M. Dalkilic, M. Kurban.
19th CPE-POWERENG, Antalya, Turkey, 2025.
- [20] **Decentralized N-1 Contingency Analysis for Cascading Failure Prediction in Multi-Region Power Systems using Consortium Blockchain.**
M. M. Islam, M. Ismail, R. Atat, **H. Kurban**, K. R. Davis, E. Serpedin.
5th International Conf. on Electrical, Computer and Energy Technologies (ICECET), Paris, France, 2025.
- [19] **Exploring Various Sequential Learning Methods for Deformation History Modeling.**
M. A. Yatkin, M. Korgesaar, J. Romanoff, J. Stuckner, Ü. Işlak, **H. Kurban**.
26th Engineering Applications of Neural Networks / Engineering Applications and Advances of Artificial Intelligence (EAAAI), Limassol, Cyprus, 2025.
- [18] **Predicting Optical Bandgaps in C₆₀ and Functionalized Derivatives from Limited Data for Renewable Energy Applications.**
M. Tuncel, **H. Kurban**[†], E. Serpedin, M. Kurban.
19th CPE-POWERENG, Antalya, Turkey, 2025.
- [17] **A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting using TimesNet** (Extended Abstract).
M. O. Cakiroglu, **H. Kurban**[†], E. K. Buxton, M. Dalkilic.
11th IEEE International Conf. on Data Science and Advanced Analytics (DSAA), San Diego, USA, 2024.
- [16] **p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data** (Extended Abstract).
P. Sharma, S. Mishra, **H. Kurban**[†], M. M. Dalkilic.
11th IEEE DSAA, San Diego, USA, 2024.
- [15] **What Data-Centric AI Can Do For k-means: A Faster, Robust k-means-d.**
P. Sharma, **H. Kurban**[†], M. M. Dalkilic.
41st International Conf. on Machine Learning (ICML, DMLR), Vienna, Austria, 2024.
- [14] **Novel NBA Fantasy League Driven by Engineered Team Chemistry and Scaled Position Statistics.**

- G. Arkanath, N. Gupta, **H. Kurban**[†], P. Sharma, M. K. R., E. K. Buxton, M. M. Dalkilic.
IEEE International Conf. on Big Data: Data-Centric AI, Sorrento, Italy, 2023.
- [13] **Are Sports Awards About Sports? Using AI to Find the Answer.**
A. Shankar, G. V. Rajasekaran, J. Hendricks, J. A. Schlak, P. Sharma, M. K. R., **H. Kurban**[†],
M. M. Dalkilic.
ECML/PKDD: 10th Workshop on Machine Learning and Data Mining for Sports Analytics,
Turin, Italy, 2023.
- [12] **ARes: An AutoML Regression Service for Data Analytics and Novel Data-centric Visualizations.**
J. Elms, S. Johnson, M. K. R., K. Sugasi, P. Sharma, **H. Kurban**[†], M. M. Dalkilic.
29th ACM SIGKDD International Conf. on Knowledge Discovery and Data Mining, Undergraduate Consortium (KDD-UC), Long Beach, CA, USA, 2023.
- [11] **Are They What They Claim: A Comprehensive Study of Ordinary Linear Regression Among the Top Machine Learning Libraries in Python.**
S. Johnson, J. Elms, M. K. R., K. Sugasi, P. Sharma, **H. Kurban**[†], M. M. Dalkilic.
29th ACM SIGKDD KDD-UC, Long Beach, CA, USA, 2023.
- [10] **Data Expressiveness and Its Use in Data-centric AI.**
H. Kurban, P. Sharma, M. M. Dalkilic.
NeurIPS Data-centric AI Workshop, 2021.
- [9] **Size Dependent Electronic Structure and Structural Properties of Cupric Oxide (CuO) Nanoparticles.**
H. Kurban, M. Kurban, M. M. Dalkilic.
International Conf. on Nanomaterials and Energy (NEM), 2019.
- [8] **Using Data Analytics to Optimize Public Transportation on a College Campus.**
K. Zimmer, **H. Kurban**, M. Jenne, L. Keating, P. Maull, M. M. Dalkilic.
IEEE DSAA, Turin, Italy, 2018.
- [7] **A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure.**
H. Kurban, M. M. Dalkilic.
IEEE International Conf. on Big Data, Boston, MA, USA, 2017.
- [6] **Improving Expectation Maximization Algorithm over Stellar Data.**
H. Kurban, C. Kockan, M. Jenne, M. M. Dalkilic.
IEEE BigData Workshop on Management, Search and Mining of Massive Repositories of Solar and Stellar Astronomy Data, Boston, MA, USA, 2017.
- [5] **Case Study: Clustering Big Stellar Data with EM*.** BEST POSTER AWARD
H. Kurban, C. Kockan, M. Jenne, M. M. Dalkilic.
IEEE/ACM International Conf. on Big Data Computing, Applications and Technologies (BDCAT),
Austin, TX, USA, 2017.
- [4] **Employing Software Engineering Principles to Enhance Management of Climatological Datasets for Coral Reef Analysis.**
M. Jenne, A. Zimmerman, **H. Kurban**, C. Johnson, M. M. Dalkilic.
6th International Workshop on Climate Informatics, Colorado, USA, 2016.

- [3] **EM*: An EM Algorithm for Big Data.** HONORABLE MENTION PAPER AWARD
H. Kurban, M. Jenne, M. M. Dalkilic.
IEEE DSAA, Montreal, Canada, 2016.
- [2] **Red-RF: Reduced Random Forest for Big Data using Priority Voting & Dynamic Data Reduction.**
 H. Mohsen, **H. Kurban**, K. Zimmer, M. Jenne, M. M. Dalkilic.
IEEE BigData Congress, New York, USA, 2015.
- [1] **A New Set of Random Forests with Varying Dynamic Data Reduction and Voting Techniques.**
 H. Mohsen, **H. Kurban**, M. Jenne, M. M. Dalkilic.
IEEE DSAA, Shanghai, China, 2014.

Book Reviews

- [3] *Data Science For All* by B. Davis & H. Glanz, 2024. ISBN: 9780135311189.
- [2] *Mastering Social Media Mining with R* by S. K. Ravindran, 2015. ISBN: 1784396311.
- [1] *Learning Data Mining with R* by B. Makhabel, 2015. ISBN: 1783982101.

OPEN-SOURCE SOFTWARE AND RESEARCH TOOLS

- **DCEM**: R package for clustering big data via data-centric modification of Expectation Maximization. Published in *SoftwareX*, 2022.
- **ccImpute**: Consensus-clustering-based imputation of dropout events in single-cell RNA-seq data; R implementation. Published in *BMC Methods*, 2022; updated version in *SoftwareX*, 2026.
- **p-ClustVal**: A novel p -adic approach for enhanced clustering and valuation in high-dimensional scRNA-seq data. Associated with *International Journal of Data Science and Analytics*, 2025.
- **Geometric- k -means**: Bound-free, energy-efficient k -means clustering algorithm. Associated with *Machine Learning*, 2025.
- **TDCM25 / CrysMTM / QuantumCanvas**: Multimodal benchmarks for temperature-dependent crystalline materials, materials foundation models, and visual learning of atomic interactions. Associated with ICLR 2025 and MLST 2025.
- **Extended de Bruijn Graph (eDBG)**: Symbolic time-series and biological-sequence representation framework. Associated with *Machine Learning: Science and Technology*, 2024, and *IEEE Access*, 2025.
- **ARes**: AutoML regression service with data-centric visualizations. Associated with ACM KDD Undergraduate Consortium, 2023.

Code repositories are linked from the Kurban Intelligence Lab website and associated paper pages.

INVITED TALKS AND PRESENTATIONS

Flagship AI/ML Conferences

- *EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration*. 39th Conf. on Neural Information Processing Systems (NeurIPS), San Diego, USA, 2025.
- *SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs*. 30th Conf. on Empirical Methods in Natural Language Processing (EMNLP) Findings, Suzhou, China, 2025.
- *Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models*. 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Towards Knowledgeable Foundation Models, Vienna, Austria, 2025.
- *Stress-Testing Multimodal Foundation Models for Crystallographic Reasoning*. 63rd ACL, Towards Knowledgeable Foundation Models, Vienna, Austria, 2025.
- *xChemAgents: Agentic AI for Explainable Quantum Chemistry*. 42nd Int. Conf. on Machine Learning (ICML), Multi-Agent Systems in the Era of Foundation Models, Vancouver, Canada, 2025.
- *Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting*. 42nd ICML, Foundation Models for Structured Data (FMSD), Vancouver, Canada, 2025.
- *PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views*. 42nd ICML, Building Physically Plausible World Models, Vancouver, Canada, 2025.
- *Beyond Atomic Geometry Representations in Materials Science: A Human-in-the-Loop Multimodal Framework*. 42nd ICML, DataWorld, Vancouver, Canada, 2025.
- *Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding*. 42nd IEEE/CVF CVPR, MM4Mat, Nashville, TN, USA, 2025. SPOTLIGHT
- *TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials*. 13th Int. Conf. on Learning Representations (ICLR), Singapore, 2025.
- *What Data-Centric AI Can Do For k-means: A Faster, Robust k-means-d*. 41st ICML, Data-centric Machine Learning Research (DMLR), Vienna, Austria, 2024.
- *p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data*. 11th IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA), San Diego, USA, 2024.
- *Novel NBA Fantasy League Driven by Engineered Team Chemistry and Scaled Position Statistics*. IEEE Int. Conf. on Big Data, Data-Centric AI, Sorrento, Italy, 2023.
- *Are Sports Awards About Sports? Using AI to Find the Answer*. ECML/PKDD 10th Workshop on Machine Learning and Data Mining for Sports Analytics, Turin, Italy, 2023.
- *Using Data Analytics to Optimize Public Transportation on a College Campus*. IEEE DSAA, Turin, Italy, 2018.
- *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure*. IEEE Int. Conf. on Big Data, Boston, MA, USA, 2017.
- *Improving Expectation Maximization Algorithm over Stellar Data*. Workshop on Management,

Search and Mining of Massive Repositories of Solar and Stellar Astronomy Data, Boston, MA, USA, 2017.

- *Case Study: Clustering Big Stellar Data with EM**. IEEE/ACM BDCAT, Austin, TX, USA, 2017.
- *EM*: An EM Algorithm for Big Data*. IEEE DSAA, Montreal, Canada, 2016.
- *A New Set of Random Forests with Varying Dynamic Data Reduction and Voting Techniques*. IEEE DSAA, Shanghai, China, 2014.

Invited Academic Colloquia and Lectures

- *Data-Centric Machine Learning*. Invited colloquia at San Jose State University, University of Houston, University of Montana, American University of the Middle East, United Arab Emirates University, University of Illinois, Zayed University, University of Missouri, and University of South Carolina, 2022 to 2023.
- *Clustering in Data Science*. Department of Statistics and Data Science, Northwestern University, 2022.
- *Comparison of Machine Learning Algorithms for CuO Nanoparticles*. 4th Int. Conf. on Physical Chemistry and Functional Materials (PCFM21), Elazığ, Turkey, 2021.
- *Practical Data Science: Examining the Correlations between Structural and Electronic Properties of Different Phases of TiO₂ Nanoparticles*. Int. Conf. on Advanced Technologies (ICAT), Istanbul, Turkey, 2020.
- *Predicting Atom Types of Anatase TiO₂ Nanoparticles with Machine Learning*. Int. Conf. on Engineering and Innovative Materials (ICEIM), Singapore, 2020.
- *Introduction to Data Science*. Computer Engineering Department, Izmir Institute of Technology, 2020.
- *Clustering Techniques in Engineering*. Computer Engineering Department, Yildirim Beyazıt University, 2019.
- *Data Science Fundamentals*. Informatics Department, Istanbul Technical University, 2018.
- *EM Algorithms for Clustering*. Computer Science Department, Indiana University, 2016.
- *Principal Component Analysis*. Computer Science Department, Indiana University, 2016.
- *Data Structures and Algorithms*. Computer Science Department, Indiana University, 2014.
- *Ensemble Models in Machine Learning*. Computer Science Department, Indiana University, 2013.

Industry, Medical, and Leadership Forums

- *AI-Driven Innovation in Medicine: Success, Failure & Challenges*. 3rd Surgical Annual Research Day, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients*. AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients*. Web Summit, Doha, Qatar, 2025.
- *A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph*. AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.

- *AI-Powered Innovation: Driving Organization*. Future Leaders Programme, Muscat, Oman, 2024.
- *Data Science and Big Data*. Eli Lilly and Company, 2017.

TEACHING EXPERIENCE

Instructor of Record

2024 to present

- **ICT 706:** Special Topics in Vision-Language Models (Graduate), Spring 2026.

2023 to present

- **ELEN & ECEN 250:** Machine Learning for Electrical Engineering (Undergraduate), Spring 2026.
- **ECEN 491:** Independent Study: Research (Undergraduate), Spring 2025, Summer 2025, Fall 2025, Spring 2026.
- **ECEN 485:** Independent Study: Directed Studies (Undergraduate), Spring 2026.
- **ENGR 110:** Introduction to Programming (Undergraduate), Fall 2024, Spring 2025, Fall 2025.
- **ENGR 102:** Engineering Lab I: Computation (Undergraduate), Spring 2024, Fall 2024, Spring 2025.
- **ECEN 210:** Computer Programming and Algorithms (Undergraduate), Fall 2023.
- **ECEN 248:** Introduction to Digital Systems Design (Undergraduate), Fall 2023.

2017 to 2023

- **CSCI-B 565:** Data Mining (Graduate), Spring 2023.
- **CSCI-B 365:** Introduction to Computers and Programming (Undergraduate), Spring 2023.
- **CSCI-C 241:** Discrete Structures for Computer Science (Undergraduate), Summer 2022, Fall 2022.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2022.
- **CSCI-B 505:** Applied Algorithms (Graduate), Fall 2021.
- **CSCI-P 556:** Applied Machine Learning (Graduate), Fall 2017.
- **INFO-I 590:** Online Applied Data Mining (Graduate), Fall 2017.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2018.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2018.

2018 to 2021

- **BMH 101:** Algorithms and Programming I (Undergraduate), Spring 2019, Fall 2019.
- **BMH 406:** Data Security (Undergraduate), Spring 2019.
- **BMH 104:** Web and Internet Technologies (Undergraduate), Spring 2019.
- **BMH 205:** Data Structures (Undergraduate), Fall 2018.
- **BMH 413:** Artificial Neural Networks (Undergraduate), Fall 2018, Fall 2019.
- **BMH 103:** Introduction to Computer Engineering (Undergraduate), Fall 2018, Fall 2019.

Associate Instructor

- **CSCI-B 565:** Data Mining (Graduate), Fall 2012, Fall 2013, Spring 2015, Spring 2016, Fall 2016.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2017.
- **CSCI-B 555:** Machine Learning (Graduate), Spring 2013.
- **CSCI-C 343:** Data Structures (Undergraduate/Graduate), Spring 2014.
- **CSCI-B 365:** Seminar in Computer Science: Data Mining (Undergraduate), Fall 2014, Fall 2015.
- **CSCI-B 609:** Topics in Algorithms and Computing Theory (Graduate), Fall 2014.
- **INFO-I 590:** Real World Data Science (Graduate, Online), Summer 2016, sponsored by Eli Lilly.

RESEARCH SUPERVISION AND MENTORING

Summary: 1 postdoctoral researcher • 1 Ph.D. graduated, 6 Ph.D. researchers in training • 6 M.Sc. researchers • 25+ undergraduates supervised.

Postdoctoral Researchers

- **Dr. Mehmet Tuncel:** completed. *Current position:* Istanbul Technical University (İTÜ), Türkiye.
Research areas: Multimodal and physics-informed machine learning; explainable AI for quantum chemistry and materials science; foundation models for scientific discovery; agentic AI; multimodal benchmarking.

*Doctoral Researchers***Graduated**

- **Dr. Parichit Sharma** (Ph.D., graduated). *Current position:* University of Maryland, Baltimore, MD, USA.
Dissertation: *Data-Centric Principles for Machine Learning: From Efficient Clustering to Single-Cell Discovery.*
Research areas: Data-centric ML; clustering theory; single-cell analytics.

Current

- **Can Polat** (Ph.D. candidate). Scientific foundation models and AI for science; multimodal and equivariant learning for atomic and crystalline systems; agentic AI for scientific reasoning and simulation; benchmark development for materials discovery and quantum chemistry.
- **Mert Onur Cakiroglu** (Ph.D. candidate). Foundation models for time-series and video; temporal reasoning and long-horizon sequence modeling; multimodal foundation models; agentic AI; symbolic representations of temporal data (multivariate de Bruijn graphs) for efficient forecasting.
- **Samir Abdaljalil** (Ph.D. candidate). Trustworthy and explainable LLMs; hallucination detection and mitigation; semantic inconsistency modeling; multilingual and code-switched evaluation; agentic reasoning frameworks (e.g., Theorem-of-Thought); sparse autoencoder-based LLM safety.

- **Md Rabiul Islam** (Ph.D. candidate). Multimodal and interpretable medical AI; lung-nodule detection and radiology-aware text fusion; Bayesian decision modeling; anatomy-aware representation learning; decision fusion under uncertainty.
- **Idil Bilge Altun** (Ph.D. student). Foundation models for sequential and dynamic data; video and time-series modeling; multimodal learning; agentic AI; hybrid symbolic–neural representations for long-horizon reasoning.
- **Rasul Khanbayov** (Ph.D. student). Vision-Language Models (VLMs); video understanding and temporal reasoning; multimodal alignment; foundation models for video and sequential data; agentic AI systems for multimodal perception and structured video analysis.

Master's Students

- **Ganesh Arkanath** (M.Sc.). *Current position:* Meta. *Thesis:* *Advancing NBA Playoffs Prediction and Fantasy Leagues Using Feature Engineering, Deep Learning, and Explainable AI*. Research areas: sports analytics; deep learning; interpretable modeling.
- **Madhavan Kalkunte Ramachandra** (M.Sc.). *Current position:* ServiceNow. Research areas: high-dimensional similarity search; k NN acceleration; telescope-based indexing; scalable ML systems.
- **Jaydeep Chauhan** (M.Sc.). *Current position:* Eli Lilly and Company. *Thesis:* *Out-of-Distribution Generalization, Spurious Correlations, and Supervised Contrastive Groupwise Learning*. Research areas: OOD generalization; debiasing; contrastive learning.
- **Alhasan Mahmood** (M.Sc.). Research areas: LLM-based evaluation; LLM-as-a-Judge frameworks; agentic AI systems.
- **Mohamed Rafi Dheen Meeran** (M.Sc.). Research areas: LLM-based evaluation; LLM-as-a-Judge frameworks; agentic AI systems.
- **Puzhuo Zheng** (M.Sc.). Research areas: computer vision; image reconstruction; representation learning; self-supervised learning.
- **Marcus Wein Monteiro** (M.Sc.). Research areas: efficient test-time reasoning for small multimodal models.

Undergraduate Researchers

- **Mohamed Rayan Barhdadi**. *Thesis:* *The Physical World as the First Teacher: Structured 4D Scene Representations*.
- **Sam Johnson**. Projects: AutoML regression benchmarking (AReS); empirical analysis of regression algorithms; data-centric diagnostics.
- **Josh Elms**. Projects: AutoML regression platform; visual analytics for model interpretability; large-scale regression behavior analysis.
- **Ahmed Al-Thani, Khalifa Al-Mansoori, Mohammed Baghazal, Ziyad Ghalayini**. Project: A Novel Extended de Bruijn Graph Framework for Hypoglycemia Prediction. **BEST CAPSTONE PROJECT AWARD**
- **Abdulaziz AlBalushi, Mohammed AlRasheed, Nasser AlHababi, Mohammed AlNaimi**. Project: Privacy-Preserving Video-Based Emotion Recognition.
- **Fadwa Abbas, Noor Alabudi, Roudha Al-Khaldi, Sandra Mansour**. *Thesis:* *A Robotics-Assisted Validation Framework for Wi-Fi-Based Indoor Localization*. **BEST CAPSTONE PROJECT**

AWARD

- **Iqra Yakub, Bashar Farooq, Ali Al-Majid, Abdellatif Hussine.** Thesis: A Privacy-Preserving Framework for Elderly Activity Monitoring Using Pose Estimation. **MULTIPLE BEST CAPSTONE AWARDS**

AWARDS

- **Noor Ahmed Al-Thani, Muneera Al-Qahtani, Aljazi Naqadan, Al-reem Al-Baker.** Thesis: An AI-Based Monitoring System for Long-Term Load Forecasting.
- **AlMaha AlEmadi, Aisha Al-Kuwari, Shahd Al-Kubaisi.** Thesis: Design and Implementation of a Web-Based Clinical Decision Support Service for Multimodal Lung Cancer Detection.

PROFESSIONAL SERVICE AND LEADERSHIP*Editorial and Advisory Roles*

- 2023 to present Editorial Board Member, *Nature Scientific Reports*
- 2023 to present Technical Editor, Manning Publications
- 2023 to 2024 Data Science Advisory Board Member, Pearson Education

Conference Organization and Program Committees

- **Shared Task Organizer:** “Ayn 2026: Cultural Grounding in Arabic Multimodal Generation and Understanding,” ArabicNLP 2026 Shared Tasks Program, co-located with EMNLP 2026, Budapest, Hungary, October 24–29, 2026.
- **Technical Program Committee Member:** IEEE International Conference on Blockchain (IEEE Blockchain), 2026.
- **Technical Program Committee Member:** IEEE Cybermatics Congress, Montbéliard, France, 2026.
- **Lead Organizer:** “AI for Science Symposium” (HBKU Horizon Hub, 1st Cycle), Doha, Qatar, January 21–22, 2026.
- **Program Committee Member:** Industry Track, The Web Conference (WWW), Dubai, UAE, 2026.
- **Invited Speaker:** Workshop on “The Role of AI in Strengthening the Resilience of Future Power Systems,” Doha, Qatar, 2025.
- **Session Chair:** 4th Digital Flow Assurance Symposium (HBKU, SLB, Wood, TAMUQ), Doha, Qatar, 2025.
- **Organizing Committee:** 19th CPE-POWERENG, Special Session on AI-Driven Innovations in Renewable Energy, 2025.
- **Organizing Committee:** 11th IEEE DSAA, Special Session on Advancing Materials Science through Data Science, 2024.
- **Program Committee Member:** Basarim High-Performance Computing Conference (2020, 2022, 2024).
- **Session Chair:** IJCAI, Jeju, South Korea, 2024.
- **Session Chair:** 11th IEEE DSAA, San Diego, USA, 2024.
- **Session Chair:** IEEE International Conference on Big Data, Sorrento, Italy, 2023.

*Peer Review***Conferences**

NeurIPS • ICML • ICLR • AAAI • ACM SIGKDD • AISTATS • COLM • IECON • Basarim High-Performance Computing Conference.

Journals

IEEE Transactions on Artificial Intelligence • *IEEE Transactions on Neural Networks and Learning Systems* • *IEEE Transactions on Circuits and Systems I: Regular Papers* • *Machine Learning: Science and Technology* • *Frontiers in Artificial Intelligence* • *Big Data Mining and Analytics* • *Journal of Chemical Information and Modeling* • *Information Sciences* • *Expert Systems with Applications* • *Engineering Applications of Artificial Intelligence* • *Applied Artificial Intelligence* • *Journal of Medical Internet Research* • *Computers in Biology and Medicine* • *Journal of Chemical Information and Modeling* • *Materials & Design* • *Applied Energy* • *Energy Conversion and Management* • *Journal of Energy Storage* • *eTransportation* • *Electric Power Systems Research* • *International Journal of Electrical Power and Energy Systems* • *Digital Communications and Networks* • *Concurrency and Computation: Practice and Experience* • *Systems Science & Control Engineering* • *Results in Engineering* • *Behaviour & Information Technology* • *International Journal of Human-Computer Interaction* • *Frontiers in Psychiatry* • *IEEE Access* • *Entertainment Computing* • *Transactions on Consumer Electronics* • *Machine Intelligence Research* • *CAAI Transactions on Intelligence Technology* • *Frontiers of Computer Science* • *Optical and Quantum Electronics* • *Frontiers in Big Data* • *SN Computer Science* • *Sustainable Energy, Grids and Networks*.

University Service

- **Hamad Bin Khalifa University** (Fall 2024 to Spring 2026): Industrial & Government Outreach Committee; ABET & Curriculum; Academic Advising; AI for Science Initiative.
- **Texas A&M University at Qatar** (Spring 2024): ABET & Curriculum; Seminars & Invited Speakers; Budget.
- **Texas A&M University at Qatar** (Fall 2023): ABET & Curriculum; Seminars & Invited Speakers; Budget; Teaching Load Model.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

2024 to present	Assistant Professor , College of Science and Engineering Hamad Bin Khalifa University, Doha, Qatar
2023 to present	Adjunct Associate Professor , Computer Science Dept. & Data Science Program Indiana University Bloomington, IN, USA
2024 to present	Adjunct Assistant Professor , Electrical & Computer Engineering Texas A&M University at Qatar, Doha, Qatar
2023 to 2024	Assistant Professor , Electrical & Computer Engineering Texas A&M University at Qatar, Doha, Qatar
2021 to 2023	Visiting Associate Professor , Computer Science Department Indiana University Bloomington, IN, USA

- 2019 to 2021 **Assistant Professor**, Computer Engineering Department
Siirt University, Siirt, Türkiye
- 2017 to 2019 **Visiting Assistant Professor**, Computer Science Department
Indiana University Bloomington, IN, USA
- 2015 to 2017 **Undergraduate Research Mentor**, Computer Science Department
Indiana University Bloomington, IN, USA
- 2015 to 2016 **Data Scientist**, Turbo Appeal, Chicago, IL, USA

EDUCATION

- 2017 **Ph.D. in Computer Science**, Minor in Statistics
Indiana University Bloomington, IN, USA
Dissertation: *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure*
Advisor: Prof. Mehmet M. Dalkilic
Committee: Prof. Predrag Radivojac • Prof. Michael W. Trosset • Prof. Yuzhen Ye

HONORS AND AWARDS

- 2026 **EDITORIAL CONTRIBUTION AWARD**: Springer Nature “Editor of Distinction Awards,” for editorial service to *Scientific Reports*
- 2025 **CVPR SPOTLIGHT**: Multimodal Learning for Materials Science (MM4Mat), Nashville, TN, USA
- 2017 **BEST POSTER AWARD**: IEEE/ACM Int. Conf. on Big Data Computing, Applications & Technologies (BDCAT), Austin, TX
- 2016 **HONORABLE MENTION PAPER AWARD**: IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA), Montreal, Canada
- 2016 to 2017 Nomination for Researcher of the Year, Indiana University Bloomington
- 2014 to 2015 Nomination for Associate Instructor of the Year, Indiana University Bloomington
- 2010 to 2012 Computer Science Graduate Fellowship, Indiana University Bloomington
- 2009 to 2017 Turkish National Ministry of Education Scholarship (comprehensive graduate studies funding)

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE) • Association for Computing Machinery (ACM).

TECHNICAL SKILLS AND LANGUAGES

Machine learning	PyTorch • JAX • TensorFlow • Hugging Face Transformers • scikit-learn • PyTorch Geometric • vLLM.
LLMs and agents	LangChain • LangGraph • LlamaIndex • OpenAI and Anthropic APIs • RAG pipelines • LoRA/QLoRA fine-tuning • LLM evaluation and red-teaming.
Systems and HPC	Python, C/C++, R, MATLAB • SLURM • distributed training (DDP, FSDP) • CUDA • Docker, Git, Linux • Weights & Biases, MLflow.
Languages	English (fluent, academic and professional) • Turkish (native).

REFERENCES

- **Prof. Mehmet M. Dalkilic:** Professor, Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington. dalkilic@indiana.edu
- **Prof. Erchin Serpedin:** Professor, Department of Electrical and Computer Engineering, Texas A&M University. eserpedin@tamu.edu
- **Prof. Hazem Nounou:** Professor, College of Science and Engineering, Hamad Bin Khalifa University. hnounou@hbku.edu.qa
- **Prof. Yuzhen Ye:** Professor, Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington. yye@indiana.edu
- **Prof. Amr Sabry:** Professor, Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington. sabry@indiana.edu
- **Prof. Michael W. Trosset:** Professor, Department of Statistics, Indiana University Bloomington. mtrosset@indiana.edu